

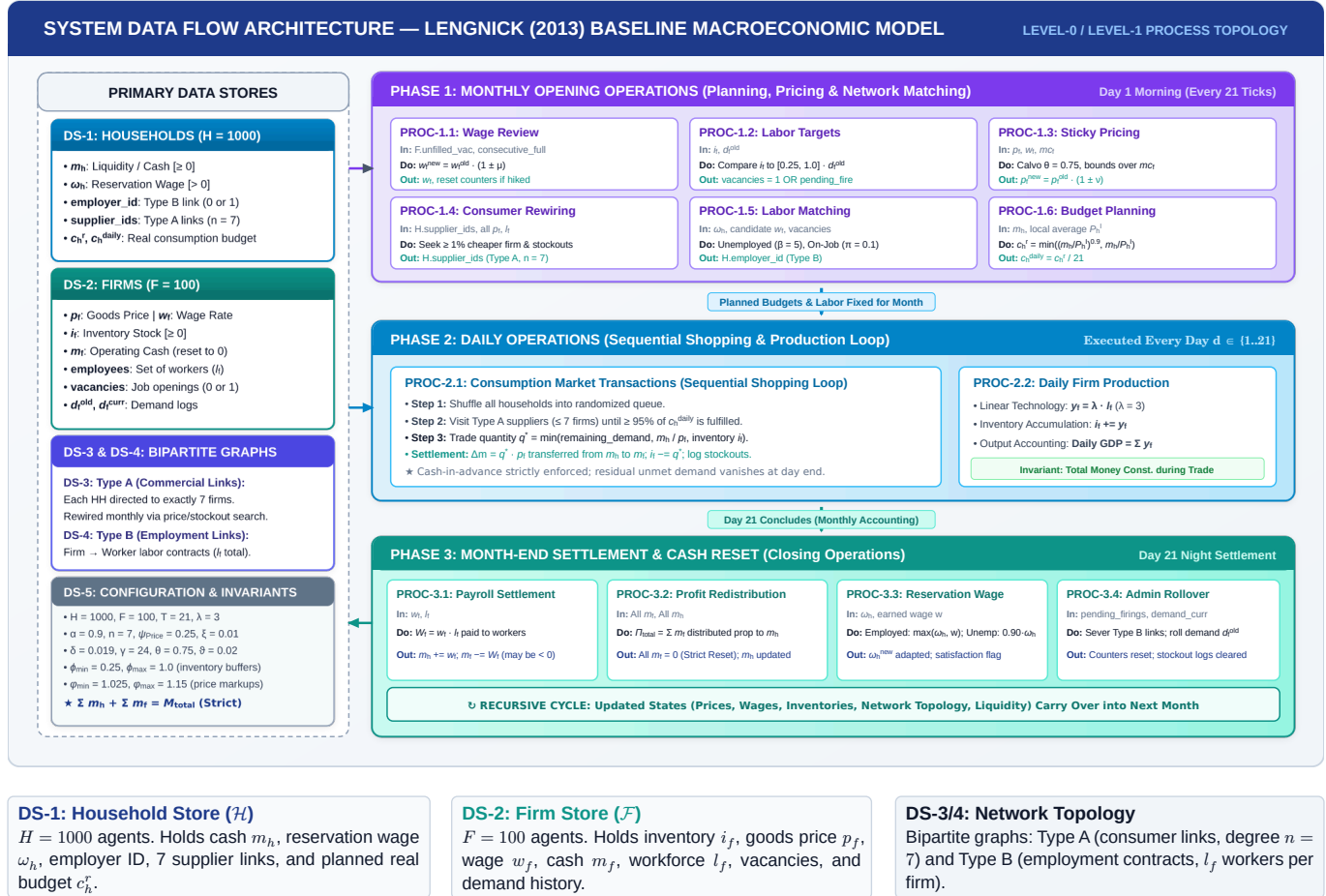
# Data Flow Architecture & Process Specifications

## Companion Implementation Guide for the Lengnick (2013) Baseline Macroeconomic Model

A complete technical blueprint mapping all data stores, entity fields, mutation semantics, and algorithmic transformations.

### 1. System-Level Data Architecture (Level-0 / Level-1 Overview)

The Lengnick model is structured into **six primary data stores** and executed through **three chronological operational phases** (Monthly Opening → Daily Trading & Production Loop → Month-End Settlement). The diagram below visualizes the global data topology, information buses, and state transitions:



## 2. Phase 1: Monthly Opening Operations (Start of Month)

Executed at the beginning of each 21-day business month ( $t \bmod 21 == 1$ ) prior to any daily transactions.

### PROC-1.1: Firm Wage Review & Adaptation

Firms evaluate past labor market success to adapt the offered wage rate  $w_f$ .

#### PROC-1.1: Firm Wage Setting

Monthly (Start) | Firm (f)

##### INPUT DATA STORES

- **DS-2:** F.unfilled\_vacancy
- **DS-2:** F.consecutive\_full
- **DS-2:** F.wage ( $w_f^{old}$ )
- **DS-5:**  $\delta = 0.019, \gamma = 24$

##### PROCESS TRANSFORMATION

Draw shock  $\mu \sim U[0, \delta]$ .  
**Shortage:** If vacancy unfilled:  
 $w_f^{new} = w_f^{old} \cdot (1 + \mu)$   
 $consecutive\_full \leftarrow 0$   
**Surplus:** Else if full  $\geq \gamma$ :  
 $w_f^{new} = w_f^{old} \cdot (1 - \mu)$   
**Neutral:** Else  $w_f^{new} = w_f^{old}$ .

##### OUTPUT MUTATIONS

- **DS-2:** F.wage ( $w_f^{new}$ )
- **DS-2:** F.consecutive\_full
- **EVENT:** WageAdapted

| Field Name                  | Store       | Type  | Description & Role                                                    |
|-----------------------------|-------------|-------|-----------------------------------------------------------------------|
| unfilled_vacancy_last_month | DS-2 (Firm) | bool  | Flag indicating if open job position went unfilled in previous month. |
| consecutive_full_months     | DS-2 (Firm) | int   | Counter of continuous months firm operated without labor deficit.     |
| wage (in)                   | DS-2 (Firm) | float | Current offered monthly wage rate before adaptation.                  |
| wage (out)                  | DS-2 (Firm) | float | Adapted monthly wage rate applied to all employees and new postings.  |
| consecutive_full (out)      | DS-2 (Firm) | int   | Reset to 0 on wage increase; preserved on cuts/neutral months.        |

### PROC-1.2: Firm Production Planning (Hiring & Firing Targets)

Firms evaluate inventory  $i_f$  against buffer thresholds derived from previous month demand  $d_f^{old}$ .

#### PROC-1.2: Labor Target Planning

Monthly (Start) | Firm (f)

##### INPUT DATA STORES

- **DS-2:** F.inventory ( $i_f$ )
- **DS-2:** F.demand\_last\_month ( $d_f^{old}$ )
- **DS-2:** F.employees ( $l_f$ )
- **DS-5:**  $\varphi = 0.25, \bar{\varphi} = 1.0$

##### PROCESS TRANSFORMATION

Compute buffer bounds:  
 $i_f^{min} = 0.25 \cdot d_f^{old}, i_f^{max} = 1.0 \cdot d_f^{old}$   
**Case 1 (Deficit,  $i_f < i_f^{min}$ ):**  
 Open 1 vacancy (vacancies = 1).  
**Case 2 (Excess,  $i_f > i_f^{max}$  &  $l_f > 0$ ):**  
 Select random worker  $h \in employees$ ;  
 Append to pending\_firings (1-month notice).  
**Case 3 (Normal Buffer):** vacancies = 0.

##### OUTPUT MUTATIONS

- **DS-2:** F.vacancies (0 or 1)
- **DS-2:** F.pending\_firings
- **DS-2:** F.inventory\_bounds
- **EVENT:** TargetPlanned

| Field Name        | Store       | Type      | Description & Role                                                |
|-------------------|-------------|-----------|-------------------------------------------------------------------|
| inventory         | DS-2 (Firm) | float     | Available finished stock at firm warehouse.                       |
| demand_last_month | DS-2 (Firm) | float     | Total units demanded by consumers in previous 21 days.            |
| vacancies         | DS-2 (Firm) | int       | Number of open jobs offered to labor market (set to 1 if hiring). |
| pending_firings   | DS-2 (Firm) | List[int] | Worker IDs scheduled for dismissal after 1-month statutory lag.   |

**PROC-1.3: Firm Calvo Sticky Price Setting**

Firms revise goods prices  $p_f$  subject to inventory pressure, marginal cost bounds, and Calvo probability  $\theta$ .

**PROC-1.3: Calvo Sticky Pricing**

Monthly (Start) | Firm (f)

**INPUT DATA STORES**

- **DS-2:** F.price ( $p_f^{old}$ )
- **DS-2:** F.wage ( $w_f$ )
- **DS-2:** F.inventory ( $i_f$ )
- **DS-2:** F.inventory\_bounds
- **DS-5:**  $\lambda = 3, \theta = 0.75, \vartheta = 0.02, \varphi_p, \bar{\varphi}_p$

**PROCESS TRANSFORMATION**

Marginal cost:  $mc_f = \frac{w_f}{21 \cdot \lambda} = \frac{w_f}{63}$   
 Bounds:  $p_f^{min} = 1.025 \cdot mc_f$ ,  $p_f^{max} = 1.15 \cdot mc_f$   
**Hike** ( $i_f < i_f^{min}$  &  $p_f < p_f^{max}$ ):  
 With prob  $\theta = 0.75$ : draw  $\nu \sim U[0, 0.02]$ ;  
 $p_f^{new} = p_f^{old} \cdot (1 + \nu)$   
**Cut** ( $i_f > i_f^{max}$  &  $p_f > p_f^{min}$ ):  
 With prob  $\theta = 0.75$ : draw  $\nu \sim U[0, 0.02]$ ;  
 $p_f^{new} = p_f^{old} \cdot (1 - \nu)$   
**Otherwise:**  $p_f^{new} = p_f^{old}$ .

**OUTPUT MUTATIONS**

- **DS-2:** F.price ( $p_f^{new}$ )
- **DS-2:** F.marginal\_cost ( $mc_f$ )
- **EVENT:** PriceAdjustment

| Field Name    | Store       | Type  | Description & Role                                                         |
|---------------|-------------|-------|----------------------------------------------------------------------------|
| wage          | DS-2 (Firm) | float | Current monthly wage determining unit marginal cost \$m c_f\$.             |
| price (in)    | DS-2 (Firm) | float | Existing selling price per unit of goods.                                  |
| marginal_cost | DS-2 (Firm) | float | Derived unit labor cost: \$w_f / (21 \cdot \lambda)\$.                     |
| price (out)   | DS-2 (Firm) | float | Updated unit selling price if adjustment condition and Calvo draw succeed. |

**PROC-1.4: Consumer Network Rewiring (Commercial Links)**

Households adapt their Type A trading network to replace expensive or stockout-prone suppliers.

**PROC-1.4: Commercial Link Rewiring**

Monthly (Start) | Household (h)

**INPUT DATA STORES**

- **DS-1:** H.supplier\_ids (Type A)
- **DS-1:** H.constrained\_firms
- **DS-2:** All F.price ( $p_f$ )
- **DS-2:** All F.employees ( $l_f$ )
- **DS-5:**  $\psi_{Price} = 0.25, \xi = 0.01, \psi_{Quant} = 0.25$

**PROCESS TRANSFORMATION**

- Price Search (Prob  $\psi_{Price} = 0.25$ ):**
  - Pick  $f_{curr} \in \text{supplier\_ids}$  uniformly at random.
  - Pick  $f_{cand} \notin \text{supplier\_ids}$  with prob  $\propto l_k$  (size).
  - If  $p_{f_{cand}} \leq (1 - \xi) \cdot p_{f_{curr}}$ :  
 Swap link: remove  $f_{curr}$ , insert  $f_{cand}$ .
- Stockout Rewiring (Prob  $\psi_{Quant} = 0.25$ ):**
  - If constrained\_firms not empty: pick  $f_{bad}$  with prob  $\propto$  unmet demand magnitude.
  - Pick random unconnected firm  $f_{new} \notin \text{supplier\_ids}$ .
  - Swap link: remove  $f_{bad}$ , insert  $f_{new}$ .

**OUTPUT MUTATIONS**

- **DS-1:** H.supplier\_ids (size 7)
- **DS-3:** Type A Adjacency Graph
- **EVENT:** SupplierReplaced

| Field Name         | Store            | Type             | Description & Role                                                                   |
|--------------------|------------------|------------------|--------------------------------------------------------------------------------------|
| supplier_ids (in)  | DS-1 (Household) | List[int]        | Existing list of 7 commercial partner firm IDs.                                      |
| constrained_firms  | DS-1 (Household) | Dict[int, float] | Map of supplier firm IDs to quantity of unmet demand in previous month.              |
| employees (all)    | DS-2 (Firm)      | Set[int]         | Workforce sizes used for preferential attachment sampling (\$P \text{ prop } l_k\$). |
| supplier_ids (out) | DS-1 (Household) | List[int]        | Updated 7-element supplier ID list preserving exact degree \$n = 7\$.                |

**PROC-1.5: Labor Market Search & Matching**

Decentralized search mechanism pairing job seekers with recruiting firms across three intensity tiers.

**PROC-1.5: Labor Search & Matching**

Monthly (Start) | Household &amp; Firm

**INPUT DATA STORES**

- **DS-1:** H.employer\_id (Type B)
- **DS-1:** H.reservation\_wage ( $\omega_h$ )
- **DS-2:** Candidate F.vacancies
- **DS-2:** Candidate F.wage ( $w_f$ )
- **DS-5:**  $\beta = 5, \pi = 0.1$

**PROCESS TRANSFORMATION**

Process households in randomized order:  
**Tier 1 (Unemployed):** Visit up to  $\beta = 5$  random firms.  
 If  $f$  has vacancies  $> 0$  and  $w_f \geq \omega_h$ :  
 $h$  hired! Set employer\_id =  $f$ ;  $f.vacancies - = 1$ ; stop.  
**Tier 2 (Employed & Unsatisfied,  $w_{empl} < \omega_h$ ):**  
 Search with prob 1.0. Visit 1 random firm  $f$ .  
 If  $f.vacancies > 0$  and  $w_f > w_{empl}$ : switch employer.  
**Tier 3 (Employed & Satisfied,  $w_{empl} \geq \omega_h$ ):**  
 Search with prob  $\pi = 0.1$ . Visit 1 random firm  $f$ .  
 If  $f.vacancies > 0$  and  $w_f > w_{empl}$ : switch employer.

**OUTPUT MUTATIONS**

- **DS-1:** H.employer\_id
- **DS-2:** F.employees
- **DS-2:** F.vacancies
- **DS-4:** Type B Adjacency Graph
- **EVENT:** Hire / JobSwitch

| Field Name       | Store            | Type       | Description & Role                                                           |
|------------------|------------------|------------|------------------------------------------------------------------------------|
| reservation_wage | DS-1 (Household) | float      | Minimum acceptable wage threshold.                                           |
| employer_id      | DS-1 (Household) | int   null | Current employer foreign key. Mutates upon hiring or job switch.             |
| vacancies        | DS-2 (Firm)      | int        | Available positions. Decrement by 1 when a match occurs.                     |
| employees        | DS-2 (Firm)      | Set[int]   | Workforce set. Household added; removed from previous employer if switching. |

**PROC-1.6: Household Consumption Budget Planning**

Households calculate their monthly real expenditure plan based on liquidity and local prices.

**PROC-1.6: Consumption Budgeting**

Monthly (Start) | Household (h)

**INPUT DATA STORES**

- **DS-1:** H.liquidity ( $m_h$ )
- **DS-1:** H.supplier\_ids
- **DS-2:** Supplier F.price ( $p_f$ )
- **DS-5:**  $\alpha = 0.9, T = 21$

**PROCESS TRANSFORMATION**

1. Calculate local average supplier price:  

$$P_h^I = \frac{1}{7} \sum_{f \in \text{supplier\_ids}_h} p_f$$
 2. Compute planned monthly real consumption:  

$$c_h^r = \min \left\{ \left( \frac{m_h}{P_h^I} \right)^\alpha, \frac{m_h}{P_h^I} \right\}, \quad \alpha = 0.9$$
 3. Determine daily real demand quota:  

$$c_h^{\text{daily}} = \frac{c_h^r}{21}$$

**OUTPUT MUTATIONS**

- **DS-1:** H.monthly\_real\_budget ( $c_h^r$ )
- **DS-1:** H.daily\_real\_demand ( $c_h^{\text{daily}}$ )
- **DS-1:** H.local\_price\_index ( $P_h^I$ )

| Field Name          | Store            | Type  | Description & Role                                         |
|---------------------|------------------|-------|------------------------------------------------------------|
| liquidity           | DS-1 (Household) | float | Available cash stock anchoring real wealth $m_h / P_h^I$ . |
| price               | DS-2 (Firm)      | float | Selling prices of the 7 connected suppliers.               |
| monthly_real_budget | DS-1 (Household) | float | Total planned real units to consume over the 21 days.      |
| daily_real_demand   | DS-1 (Household) | float | Daily consumption target quota: $c_h^r / 21$ .             |

### 3. Phase 2: Daily Operations Processes (Days 1 to 21)

Executed each business day  $d \in \{1, \dots, 21\}$  in strict sequence: Consumption Market  $\rightarrow$  Production Process.

#### PROC-2.1: Consumption Market Transactions (Sequential Shopping)

Households visit suppliers to buy goods under cash-in-advance and inventory constraints.

##### PROC-2.1: Daily Shopping Transactions

Daily (Days 1..21) | Household & Firm

###### INPUT DATA STORES

- **DS-1:** H.daily\_real\_demand ( $c_h^{daily}$ )
- **DS-1:** H.liquidity ( $m_h$ )
- **DS-1:** H.supplier\_ids
- **DS-2:** Visited F.price ( $p_f$ )
- **DS-2:** Visited F.inventory ( $i_f$ )
- **DS-2:** Visited F.liquidity ( $m_f$ )

###### PROCESS TRANSFORMATION

Initialize  $q_{rem} = c_h^{daily}$ ,  $k = 0$ .  
**While**  $k < 7$  **and**  $q_{rem} > 0.05 \cdot c_h^{daily}$ :  
 • Pick unvisited firm  $f \in \text{supplier\_ids}$ ;  $k \leftarrow k + 1$ .  
 • Max affordable:  $q_{aff} = \frac{m_h}{p_f}$ ; Available:  $q_{avail} = i_f$ .  
 • Desired purchase:  $q_{des} = \min(q_{rem}, q_{aff})$ .  
 • Executed quantity:  $q^* = \min(q_{des}, q_{avail})$ .  
 • Log demand:  $f.\text{demand\_current\_month} += q_{des}$ .  
 • Log stockout: **if**  $q^* < q_{des}$ , **log unmet** in  $h.\text{constrained\_firms}$ .  
 • Money/goods transfer:  
 $m_h \leftarrow m_h - q^* \cdot p_f$ ,  $m_f \leftarrow m_f + q^* \cdot p_f$   
 $i_f \leftarrow i_f - q^*$ ,  $q_{rem} \leftarrow q_{rem} - q^*$

###### OUTPUT MUTATIONS

- **DS-1:** H.liquidity ( $m_h$ )
- **DS-1:** H.constrained\_firms
- **DS-2:** F.liquidity ( $m_f$ )
- **DS-2:** F.inventory ( $i_f$ )
- **DS-2:** F.demand\_current\_month
- **EVENT:** GoodPurchased

| Field Name           | Store            | Type             | Description & Role                                                                |
|----------------------|------------------|------------------|-----------------------------------------------------------------------------------|
| liquidity (HH)       | DS-1 (Household) | float            | Decrement by purchase expenditure $q^* \cdot p_f$ . Strictly non-negative.        |
| constrained_firms    | DS-1 (Household) | Dict[int, float] | Logs quantity of unmet demand at firm $f$ for end-of-month rewiring.              |
| liquidity (Firm)     | DS-2 (Firm)      | float            | Incremented by sales revenue $q^* \cdot p_f$ . Accumulates daily.                 |
| inventory            | DS-2 (Firm)      | float            | Decrement by physical units sold $q^*$ .                                          |
| demand_current_month | DS-2 (Firm)      | float            | Accumulates customer order intentions $q_{des}$ regardless of stock availability. |

#### PROC-2.2: Daily Production Process

Firms utilize employed labor to manufacture consumption goods.

##### PROC-2.2: Daily Firm Production

Daily (Days 1..21) | Firm (f)

###### INPUT DATA STORES

- **DS-2:** F.employees ( $l_f$ )
- **DS-2:** F.inventory ( $i_f^{old}$ )
- **DS-5:**  $\lambda = 3$

###### PROCESS TRANSFORMATION

For each firm  $f \in \{1, \dots, F\}$  on day  $d$ :  
 1. Calculate output using linear labor technology:  
 $y_f = \lambda \cdot l_f = 3 \cdot l_f$   
 2. Add produced output to warehouse inventory:  
 $i_f^{new} = i_f^{old} + y_f$

###### OUTPUT MUTATIONS

- **DS-2:** F.inventory ( $i_f^{new}$ )
- **DS-6:** METRICS.daily\_aggregate\_gdp
- **EVENT:** ProductionCompleted

| Field Name          | Store          | Type     | Description & Role                                                      |
|---------------------|----------------|----------|-------------------------------------------------------------------------|
| employees           | DS-2 (Firm)    | Set[int] | Workforce size $l_f =  \text{"employees"} $ governing daily production. |
| inventory (in)      | DS-2 (Firm)    | float    | Warehouse stock prior to daily production batch.                        |
| inventory (out)     | DS-2 (Firm)    | float    | Warehouse stock after adding new output: $i_f + 3 \cdot l_f$ .          |
| daily_aggregate_gdp | DS-6 (Metrics) | float    | Global daily output tracker: $\sum_f y_f$ .                             |

#### 4. Phase 3: Month-End Settlement & Accounting (Day 21 Close)

Executed at the conclusion of day 21. Handles payroll, profit distribution, cash resets, and reservation wage updates.

##### PROC-3.1: Payroll & Wage Disbursement

Firms disburse contractual wages to their workforce.

##### PROC-3.1: Monthly Wage Settlement

Month-End (Day 21) | Firm & Household

###### INPUT DATA STORES

- **DS-2:** F.wage ( $w_f$ )
- **DS-2:** F.employees ( $l_f$ )
- **DS-2:** F.liquidity ( $m_f^{old}$ )
- **DS-1:** Employee H.liquidity ( $m_h^{old}$ )

###### PROCESS TRANSFORMATION

For each firm  $f \in \{1, \dots, F\}$ :

- Calculate total payroll obligation:  $W_f = w_f \cdot l_f$ .
- Deduct payroll from firm cash:  $m_f \leftarrow m_f - W_f$ .
- For each worker  $h \in \text{employees}$ :  
Transfer wage:  $m_h \leftarrow m_h + w_f$ .

**(Note: Firm liquidity  $m_f$  can become negative if payroll exceeds sales revenue).**

###### OUTPUT MUTATIONS

- **DS-1:** Employee H.liquidity ( $m_h^{new}$ )
- **DS-2:** F.liquidity ( $m_f^{new}$ )
- **EVENT:** WagesPaid

| Field Name       | Store            | Type  | Description & Role                                                        |
|------------------|------------------|-------|---------------------------------------------------------------------------|
| wage             | DS-2 (Firm)      | float | Fixed monthly wage rate paid to each employed worker.                     |
| liquidity (Firm) | DS-2 (Firm)      | float | Operating cash balance decremented by total wage bill $\$w_f \cdot l_f$ . |
| liquidity (HH)   | DS-1 (Household) | float | Employee cash balance incremented by monthly wage $\$w_f$ .               |

##### PROC-3.2: Profit & Deficit Redistribution (Firm Cash Reset)

All remaining operational cash (profits) or deficits across firms are cleared to households proportional to household liquidity.

##### PROC-3.2: Profit/Deficit Redistribution

Month-End (Day 21) | All Firms & Households

###### INPUT DATA STORES

- **DS-2:** All F.liquidity ( $m_f$  post-pay-roll)
- **DS-1:** All H.liquidity ( $m_h$  post-pay-roll)

###### PROCESS TRANSFORMATION

1. Compute aggregate firm operating balance:  
 $\Pi_{\text{total}} = \sum_{f=1}^F m_f$
2. Compute aggregate household cash balance:  
 $M_{\text{total}}^{HH} = \sum_{h=1}^H m_h$
3. Distribute dividend/deficit shares to each household:  
 $\Delta m_h = \Pi_{\text{total}} \cdot \left( \frac{m_h}{M_{\text{total}}^{HH}} \right)$   
 $m_h \leftarrow m_h + \Delta m_h$
4. Reset all firm cash balances to zero:  
 $m_f \leftarrow 0 \quad \forall f \in \{1, \dots, F\}$

###### OUTPUT MUTATIONS

- **DS-1:** All H.liquidity ( $m_h$  updated)
- **DS-2:** All F.liquidity ( $m_f = 0$  strictly)
- **DS-6:** METRICS.aggregate\_profits
- **EVENT:** DividendsDistributed

| Field Name           | Store            | Type  | Description & Role                                                             |
|----------------------|------------------|-------|--------------------------------------------------------------------------------|
| liquidity (Firm in)  | DS-2 (Firm)      | float | Net post-wage operational cash balance $\$Pi_f = m_f$ .                        |
| liquidity (HH in)    | DS-1 (Household) | float | Household cash holding determining dividend ownership share $\$m_h / M^{HH}$ . |
| liquidity (HH out)   | DS-1 (Household) | float | Updated household cash after dividend/deficit allocation.                      |
| liquidity (Firm out) | DS-2 (Firm)      | float | Identically reset to 0.0 for every firm. Guaranteed by cash reset invariant.   |

**PROC-3.3: Household Reservation Wage Update**

Households adjust their reservation wage  $\omega_h$  based on labor income received during the month.

**PROC-3.3: Reservation Wage Adaptation**

Month-End (Day 21) | Household (h)

**INPUT DATA STORES**

- **DS-1:** H.reservation\_wage ( $\omega_h^{old}$ )
- **DS-1:** H.employer\_id
- **DS-2:** Employer F.wage ( $w_f$ )
- **DS-5:**  $\Delta\omega = 0.10$

**PROCESS TRANSFORMATION****Case 1 (Employed Worker):**

Let wage received be  $w_{rec} = w_{employer}$ .

- If  $w_{rec} > \omega_h^{old}$ : ratchet up  $\omega_h^{new} = w_{rec}$ .

- If  $w_{rec} \leq \omega_h^{old}$ : keep  $\omega_h^{new} = \omega_h^{old}$   
(worker becomes unsatisfied; searches with prob 1.0 next month).

**Case 2 (Unemployed throughout month):**

Apply 10% statutory markdown:

$$\omega_h^{new} = \omega_h^{old} \cdot (1 - 0.10) = 0.90 \cdot \omega_h^{old}$$

**OUTPUT MUTATIONS**

- **DS-1:** H.reservation\_wage ( $\omega_h^{new}$ )
- **DS-1:** H.is\_satisfied
- **EVENT:** ReservationWageAdapted

| Field Name             | Store            | Type       | Description & Role                                                                  |
|------------------------|------------------|------------|-------------------------------------------------------------------------------------|
| reservation_wage (in)  | DS-1 (Household) | float      | Previous claim on monthly wage income.                                              |
| employer_id            | DS-1 (Household) | int   null | Identifies employment state (null = unemployed).                                    |
| wage                   | DS-2 (Firm)      | float      | Contractual wage earned from employer.                                              |
| reservation_wage (out) | DS-1 (Household) | float      | Updated reservation wage claim for next month's matching.                           |
| is_satisfied           | DS-1 (Household) | bool       | Derived status: $\$w(e m p l) \geq \omega_h$ . Directs on-the-job search intensity. |

**PROC-3.4: Administrative Rollover & Deferred Termination Execution**

System cleans temporary registers, executes delayed worker dismissals, updates historical counters, and rolls forward time.

**PROC-3.4: Administrative Rollover**

Month-End (Day 21) | Global / All Agents

**INPUT DATA STORES**

- **DS-2:** F.pending\_firings
- **DS-2:** F.vacancies
- **DS-2:** F.demand\_current\_month
- **DS-1:** H.constrained\_firms

**PROCESS TRANSFORMATION****1. Execute Deferred Firings:**

For each  $h \in \text{pending\_firings}$  of firm  $f$ :

- Remove  $h$  from  $f$ .employees.
- Set  $h$ .employer\_id = null (Type B link severed).
- Clear pending\_firings list.

**2. Update Staffing Continuity Counters:**

- If  $f$ .vacancies > 0 (unfilled position):

Set unfilled\_vacancy\_last\_month = true,  
consecutive\_full\_months = 0.

- Else if  $f$ .vacancies == 0 and  $l_f > 0$ :

Set unfilled\_vacancy\_last\_month = false,  
consecutive\_full\_months += 1.

- Reset open vacancies = 0.

**3. Rollover Demand Registers:**

- $d_f^{old} \leftarrow \text{demand\_current\_month}$  for firm  $f$ .

- demand\_current\_month for firm  $f \leftarrow 0$ .

**4. Clear Consumer Stockout Logs:**

- Reset  $h$ .constrained\_firms = {} for all households.

**OUTPUT MUTATIONS**

- **DS-1:** H.employer\_id ("null" if fired)
- **DS-1:** H.constrained\_firms (cleared)
- **DS-2:** F.employees (reduced)
- **DS-2:** F.pending\_firings (cleared)
- **DS-2:** F.demand\_last\_month ( $d_f^{old}$ )
- **DS-2:** F.consecutive\_full\_months
- **DS-4:** Type B Adjacency Graph
- **EVENT:** FiringsExecuted / MonthRolled

| Field Name           | Store            | Type             | Description & Role                                                                      |
|----------------------|------------------|------------------|-----------------------------------------------------------------------------------------|
| pending_firings      | DS-2 (Firm)      | List[int]        | Worker IDs discharged now after the 1-month job-protection delay.                       |
| demand_current_month | DS-2 (Firm)      | float            | Total daily demand accumulated over the month; copied into $\$d\_f(o l d)$ then zeroed. |
| demand_last_month    | DS-2 (Firm)      | float            | Updated baseline demand value used to anchor inventory buffers next month.              |
| constrained_firms    | DS-1 (Household) | Dict[int, float] | Emptied to start fresh for upcoming month's transactions.                               |

5. Entity-Process Mutation Matrix (CRUD Matrix)

This matrix maps every entity field to the processes that **C**\*reate, **R**\*ead, **U**\*pdate, or **D**\*elete it. It provides an immediate dependency map for structuring state machines and actor architectures:

| Entity Field                       | P-1.1 | P-1.2 | P-1.3 | P-1.4 | P-1.5 | P-1.6 | P-2.1 | P-2.2 | P-3.1 | P-3.2 | P-3.3 |
|------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| HH.liquidity ( $m_h$ )             | -     | -     | -     | -     | -     | R     | U     | -     | U     | U     | -     |
| HH.reservation_wage ( $\omega_h$ ) | -     | -     | -     | -     | R     | -     | -     | -     | -     | -     | U     |
| HH.employer_id (Type B)            | -     | -     | -     | -     | U     | -     | -     | -     | R     | -     | R     |
| HH.supplier_ids (Type A)           | -     | -     | -     | U     | -     | R     | R     | -     | -     | -     | -     |
| HH.monthly_real_budget             | -     | -     | -     | -     | -     | U     | R     | -     | -     | -     | -     |
| HH.daily_real_demand               | -     | -     | -     | -     | -     | U     | R     | -     | -     | -     | -     |
| HH.constrained_firms               | -     | -     | -     | R     | -     | -     | U     | -     | -     | -     | -     |
| Firm.wage ( $w_f$ )                | U     | -     | R     | -     | R     | -     | -     | -     | R     | -     | R     |
| Firm.price ( $p_f$ )               | -     | -     | U     | R     | -     | R     | R     | -     | -     | -     | -     |
| Firm.inventory ( $i_f$ )           | -     | R     | R     | -     | -     | -     | U     | U     | -     | -     | -     |
| Firm.liquidity ( $m_f$ )           | -     | -     | -     | -     | -     | -     | U     | -     | U     | U     | -     |
| Firm.employees ( $l_f$ )           | -     | R     | -     | R     | U     | -     | -     | R     | R     | -     | -     |
| Firm.vacancies                     | -     | U     | -     | -     | U     | -     | -     | -     | -     | -     | -     |
| Firm.pending_firings               | -     | U     | -     | -     | -     | -     | -     | -     | -     | -     | -     |
| Firm.demand_last_month             | -     | R     | R     | -     | -     | -     | -     | -     | -     | -     | -     |
| Firm.demand_current                | -     | -     | -     | -     | -     | -     | U     | -     | -     | -     | -     |
| Firm.consecutive_full              | U     | -     | -     | -     | -     | -     | -     | -     | -     | -     | -     |

6. Simulation Engine Orchestration Loop

The high-level execution algorithm binding all 10 processes across daily and monthly cycles:

```
for month in range(1, TOTAL_MONTHS + 1):
    # PHASE 1: MONTHLY OPENING (Planning, Pricing, Network Matching)
    run_proc_1_1_firm_wage_setting(firms, cfg)
    run_proc_1_2_firm_labor_planning(firms, cfg)
    run_proc_1_3_firm_price_setting(firms, cfg)
    run_proc_1_4_consumer_network_rewiring(households, firms, cfg)
    run_proc_1_5_labor_market_matching(households, firms, cfg)
    run_proc_1_6_household_consumption_budgeting(households, firms, cfg)

    # PHASE 2: DAILY TRADING & PRODUCTION (21 Business Days)
    for day in range(1, 22):
        run_proc_2_1_daily_shopping_transactions(households, firms, cfg)
        run_proc_2_2_daily_firm_production(firms, cfg)
        assert_money_conservation(households, firms, total_money)

    # PHASE 3: MONTH-END SETTLEMENT & REVISION
    run_proc_3_1_monthly_wage_settlement(firms, households)
    run_proc_3_2_profit_deficit_redistribution(firms, households)
    run_proc_3_3_reservation_wage_adaptation(households, firms, cfg)
    run_proc_3_4_administrative_rollover(firms, households)
    assert_firm_cash_cleared(firms)
```

7. Data Integrity Assertions & Simulation Invariant Checks

Place these explicit assertion checks directly into your simulation engine loop to guarantee mathematical and behavioral integrity:

- 1. **Global Money Conservation Invariant (Verify at every daily and monthly tick  $t$ ):**  
 $\left| \left( \sum_{h=1}^H m_h + \sum_{f=1}^F m_f \right) - M_{\text{total}} \right| < 10^{-7}$   
Assertion failure indicates cash leakage during shopping transactions, payroll disbursements, or dividend allocations.
- 2. **Firm Solvency / Cash-Clearing Invariant (Verify at day 1 morning of each month):**  
 $m_f == 0.0 \quad \forall f \in \{1, \dots, F\}$   
Firms retain zero cash balances across months; all operational earnings/deficits are cleared via PROC-3.2.
- 3. **Household Budget Invariant (Verify before every shopping visit in PROC-2.1):**  
 $m_h \geq 0.0 \quad \forall h \in \{1, \dots, H\}$   
Households cannot run overdrafts; purchases are strictly bounded by cash-in-advance ( $q^* \leq \frac{m_h}{p_f}$ ).
- 4. **Network Topology Invariant (Verify after PROC-1.4 rewiring):**  
 $|\text{supplier\_ids}_h| == 7 \quad \forall h \in \{1, \dots, H\}$   
Every household must preserve exactly  $n = 7$  active Type A commercial connections.
- 5. **Labor Force Conservation Invariant (Verify after PROC-1.5 matching and PROC-3.4 dismissals):**  
 $\sum_{f=1}^F |\text{employees}_f| + |\{h : \text{employer\_id}_h == \text{null}\}| == H = 1000$   
No worker can be employed by multiple firms simultaneously or vanish from the labor accounting register.